

AI-Powered Intrusion Detection Systems: Enhancing Real-Time Network Threat Monitoring - A Systematic Review

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Abstract

Network security is paramount as modern IT infrastructures face increasingly sophisticated threats. Traditional Intrusion Detection Systems (IDS) struggle to cope with evolving attack patterns and high-speed network traffic. Artificial Intelligence (AI) has emerged as a transformative force, enabling real-time, adaptable, and accurate intrusion detection. This systematic review explores the state-of-the-art applications of AI in IDS, with a focus on real-time threat monitoring. We analyze methodologies including machine learning, deep learning, and hybrid approaches, discuss current challenges, and present future directions.

Keywords: Intrusion Detection Systems, Artificial Intelligence, Real-Time Monitoring, Network Security, Machine Learning

1. Introduction

With the proliferation of digital technologies and connectivity, networks have become increasingly vulnerable to cyberattacks (Li et al., 2020). Intrusion Detection Systems (IDS) are critical for protecting networked systems by identifying unauthorized access and malicious activities (Sommer & Paxson, 2010). However, traditional IDS techniques, reliant on signature-based or manual rule-based detection, fail to detect new or obfuscated threats and struggle with high-speed, high-volume data streams.

Artificial Intelligence (AI), especially Machine Learning (ML) and Deep Learning (DL), is revolutionizing IDS, promising enhanced detection accuracy, adaptability, and real-time threat monitoring (Buczak & Guven, 2016). This review systematically surveys the literature on AI-powered IDS, their methodologies, performance in real-time network environments, challenges, and research gaps.

2. Methodology

2.1 Search Strategy

The review follows PRISMA guidelines, focusing on articles published between 2015 and 2024 indexed in IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink. Keywords included “AI-powered IDS”, “intrusion detection”, “machine learning IDS”, “real-time network security”, and “deep learning intrusion detection.”

2.2 Inclusion and Exclusion Criteria

Studies were included if they:

- Focused on the use of AI/ML/DL in IDS,
- Addressed real-time or online detection,
- Provided performance evaluation against benchmark datasets, and
- Were peer-reviewed full articles in English.

Exclusions:

- Studies focused exclusively on signature or rule-based IDS,
- Offline-only detection without real-time applicability,
- Abstracts, patents, and non-peer-reviewed materials.

2.3 Data Extraction and Synthesis

A total of 1480 articles were identified, with 28 meeting inclusion criteria. Key data were extracted into comparative tables regarding AI techniques, datasets used, detection performance, and limitations.

3. Background: Intrusion Detection Systems

3.1 IDS Types and Architectures

Intrusion Detection Systems are classified as (1) Signature-based, (2) Anomaly-based, and (3) Hybrid. Signature-based IDS, like Snort, detect known threats but miss zero-day attacks. Anomaly-based IDS, often powered by AI, detect deviations from norm and excel at spotting novel threats (Garcia-Teodoro et al., 2009).

Architecturally, IDS are divided into Host-based IDS (HIDS) and Network-based IDS (NIDS). NIDS focus on monitoring network traffic in real time, making them crucial for enterprise security (Scarfone & Mell, 2007).

3.2 Limitations of Conventional IDS

Conventional IDS face challenges:

- Low detection rates for unknown/new attacks
- High false positives
- Scalability and real-time bottlenecks
- Inability to adapt to changing attack patterns

These limitations motivate the adoption of AI to enhance intrusion detection capabilities.

4. AI Techniques in Intrusion Detection

4.1 Machine Learning Approaches

4.1.1 Supervised Learning

Supervised ML methods, like Decision Trees, Support Vector Machines (SVM), and Random Forests, are trained on labeled data to classify traffic as benign or malicious (Tsai et al., 2009). Works such as Moustafa & Slay (2016) highlight that Random Forest and SVM can outperform classical methods, especially on complex datasets like UNSW-NB15.

4.1.2 Unsupervised Learning

Unsupervised methods, such as K-means clustering and autoencoders, detect anomalous patterns in unlabeled data, crucial for unknown attacks (Shone et al., 2018).

4.1.3 Ensemble Methods

Ensemble methods combine multiple models to improve robustness, e.g., mixing tree-based and neural models (Kim et al., 2014).

4.2 Deep Learning Approaches

Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN, especially Long Short-Term Memory - LSTM) have shown promise for sequence-aware and scalable IDS (Krizhevsky et al., 2012; Yin et al., 2017).

CNNs excel at extracting spatial features from packet data, while LSTMs capture temporal dependencies in network flows.

4.3 Reinforcement Learning and Hybrid Models

Reinforcement Learning (RL) enables IDS to learn optimal detection policies in dynamic environments (Nguyen & Reddi, 2019). Hybrid models, such as combining ML and signature techniques, provide balance between novelty detection and efficiency (Javaid et al., 2016).

5. Datasets and Benchmarking

Effective evaluation of IDS relies on realistic datasets. Well-known datasets include:

- KDD Cup 99: A classic but criticized for redundancy and unrealistic attack distribution (Tavallae et al., 2009).
- NSL-KDD: An enhanced KDD version with better evaluation properties (Mahmood et al., 2013).
- UNSW-NB15: Includes contemporary attack types and normal modern traffic (Moustafa & Slay, 2015).
- CICIDS 2017/2018: Realistic up-to-date network traffic with up-to-date attack vectors (Sharafaldin et al., 2018).

Many recent studies recommend UNSW-NB15 and CICIDS as preferable due to representativeness of modern network environments.

6. Review of Recent Advances (2015-2024)

6.1 Machine Learning-based IDS

Moustafa & Slay (2016) applied statistical feature selection and ensemble classifiers to UNSW-NB15, achieving high accuracy (over 85%) with reduced false positives. Similarly, Buczak & Guven (2016) surveyed multiple ML techniques, noting Decision Trees and Random Forests dominated due to speed and interpretability.

He et al. (2020) applied SVM with feature engineering for NIDS, attaining 97.5% accuracy on the CICIDS 2017 dataset.

6.2 Deep Learning-based IDS

Yin et al. (2017) introduced LSTM networks for anomaly-based IDS, improving detection rates for both known and unknown attacks. Shone et al. (2018) leveraged deep autoencoders for feature extraction, followed by shallow classifiers, enhancing performance on the NSL-KDD dataset.

Kim et al. (2019) proposed a CNN-LSTM hybrid, capturing both packet features and temporal context, and tested it on CICIDS 2017 and UNSW-NB15, outperforming baselines.

6.3 Online and Real-time Detection

A common challenge is translating academic performance to real-time monitoring. Lopez-Martin et al. (2017) built an online deep learning NIDS based on incremental learning, capable of adapting to evolving attack patterns. Their model maintained high throughput and adaptability.

Bhuyan et al. (2014) reviewed scalable real-time NIDS and found that stream-based learning models (incremental updates, resource awareness) are more effective in practical deployments.

7. Performance Evaluation

7.1 Metrics

IDS are evaluated using:

- Accuracy
- Precision, Recall, F1-Score
- False Positive Rate (FPR)
- Area Under ROC Curve (AUC)
- Detection Latency

7.2 Comparative Analysis

Table 1 summarizes top AI-powered IDS:

Reference	Method	Dataset	Accuracy	FPR	Real-Time Capable
Moustafa & Slay (2016)	RF, SVM	UNSW-NB15	85-90%	<2%	Yes
Yin et al. (2017)	LSTM	NSL-KDD	95.70%	2.48%	Partially
Kim et al. (2019)	CNN-LSTM	CICIDS 2017	98.10%	1.60%	Yes
Shone et al. (2018)	Autoencoder	NSL-KDD	87.10%	3.80%	No
Lopez-Martin et al. (2017)	Deep Online	Custom	90%+	-	Yes

8. Challenges and Limitations

8.1 Data Quality and Availability

Most public datasets are outdated or do not reflect modern encrypted traffic patterns (Ring et al., 2019).

8.2 Imbalanced Data

Attack samples are often underrepresented, leading to poor detection of rare (zero-day) attacks (Maloof, 2006).

8.3 Adversarial Attacks

AI models are susceptible to adversarial samples—carefully crafted packets that evade detection (Hu & Tan, 2017).

8.4 Scalability and Latency

Deep learning models, while accurate, may incur non-negligible processing latency, challenging for gigabit/terabit networks (Bailey et al., 2014).

8.5 Explainability

Black-box nature of many deep learning IDS makes it difficult to interpret alerts or justify decisions (Xie et al., 2021).

9. Future Directions

- Adversarial Robustness: Enhancing models to resist evasion techniques.
- Explainable AI: Building interpretable models for operational trust.
- Federated Learning: Collaborative IDS models without data sharing for privacy (Nguyen et al., 2022).
- Online/Incremental Learning: Adapting models as new attack types emerge (Almseidin et al., 2017).
- Integration with SDN: Leveraging software-defined networking for dynamic network defense (Zhou et al., 2020).
- Dataset Creation: Developing realistic, up-to-date datasets reflecting current network and attack trends.

10. Conclusion

AI-powered IDS represent the future of network security, delivering substantial improvements in real-time threat detection and adaptability. Machine learning and deep learning techniques have already demonstrated superior accuracy and lower false positives compared to traditional methods. However, practical deployment faces challenges including adversarial risks, scalability, and explainability. Ongoing research into online learning, adversarial defense, and integration with new networking paradigms (e.g., SDN, IoT) will further enhance AI-driven IDS utility for real-time network cybersecurity.

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